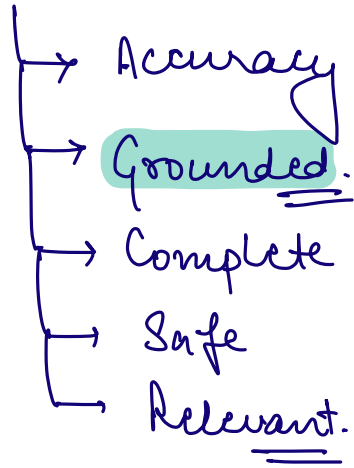


Goal of a good RAG application.



Evaluation framework of RAG.

- right document retrieved?
- right chunk retrieved?
- final response correct?
- Grounded response?

Prepare test questions



Sample Policy documents.



Run RAG Application



Response.

Sample Evaluation Dataset for E-Commerce RAG

Test Case	Customer Query	Expected Policy	Expected Answer Points
1	Can I return shoes after 15 days?	Return	Return allowed within 30 days if unused and original packaging
2	Can I return opened personal care items?	Return	Opened personal care items are not returnable
3	How long will my COD refund take?	Refund	Bank details required; 7–10 business days
4	Are shipping charges refundable?	Refund	Non-refundable unless damaged, defective, or incorrectly delivered
5	How long does express shipping take?	Shipping	1–2 business days, extra charges may apply
6	Is standard shipping free?	Shipping	Free for orders above ₹999
7	Does warranty cover accidental damage?	Warranty	No, accidental damage is not covered
8	What do I need for warranty claim?	Warranty	Invoice, serial number, images/videos may be required

Use a 0–2 scale.

Score	Meaning
0	Retrieved content is irrelevant
1	Partially relevant but missing important information
2	Correct and complete context retrieved

Example:

Query	Retrieved Content	Score
COD refund timeline	COD refunds take 7–10 business days	2
COD refund timeline	Refunds processed after quality check only	1
COD refund timeline	Shipping takes 3–5 days	0

Retrieval Metrics.

→ Precision: Out of retrieved documents | chunks, how many are actually relevant?

Retrieved documents $\Rightarrow$ ④

Relevant documents $\Rightarrow$ ②

$$\text{Precision} = \frac{2}{4} = 0.5 \Rightarrow \underline{\underline{50\%}}$$

High Precision means less Noise

↓
Irrelevant
Documents / Chunks

Recall.

↳ Out of all the relevant documents, how many did we retrieve?

Ex: let's say 4 relevant docs are there in the knowledge.

Retrieved = 3

$$\text{Recall} = \frac{3}{4} = 0.75$$

Precision (vs) Recall.

↓
Avoid irrelevant
documents.

↘ Avoid missing
relevant document.

Top-K Recall.

↳ Check whether the correct answer appears within the top-k retrieved documents or not.

Example:

Query:
How long does COD refund take?

Top-1 result:
General refund **policy**

Top-2 result:
COD refund policy

Here:

Recall@1 = Failed
Recall@2 = Passed

evaluate response accuracy

Use a 0–3 scale.

Score	Meaning
0	Wrong answer
1	Partially correct but major information missing
2	Mostly correct but minor issue
3	Fully correct and grounded 

Example:

Question:
Does warranty cover liquid damage?

Policy:
Liquid damage **is not** covered.

Answer 1:
Yes, warranty covers liquid damage.
Score: 0

Answer 2:
No, warranty **does not** cover liquid damage.
Score: 3

Answer 3:
No, warranty **does not** cover liquid damage, **but** you may **get** a free replacement.
Score: 1 or 2 depending on whether replacement **is** unsupported.

Detect Hallucination in Responses.

Improve grounding in response

→ Answer should be based
only on the retrieved documents.

→ No random assumptions in the answer.

Time Top-K parameter

Apply metadata filtering for better precision

Improve prompt design for policy based Answer.

Weak Prompt

Answer the **customer** question.

Problem:

The model may **use general** knowledge **or guess**.

Strong Prompt

You are a **customer** support assistant **for** an e-commerce company.

Use only the **policy** context provided below.

Rules:

1. **Do not** make up **policy** details.
2. **If** the answer is **not** present **in** the context, say you **do not** have enough information.
3. Mention timelines, eligibility rules, **and** exceptions **if** present.
4. Keep the answer clear **and** customer-friendly.

Continuous Evaluation & Improvement Process.